

STAT 301 - Statistical Modelling for Data Science

(2026, Summer Term 2)

Lecture:

Tuesday/Thursday 2pm-5pm, ESB1012. July 7 - August 13, *in-person*.

Tutorial:

Thursdays 12:30pm–1:30pm, ESB1012.

Class webpage:

UBC Canvas STAT301.

Instructor:

Lang Wu, Professor, Department of Statistics, UBC.

Personal webpage: <https://www.stat.ubc.ca/~lang>

Email: lang@stat.ubc.ca

Office: *Earth Science Building (ESB) 3126*, 2207 Main Mall.

Office hours:

See Canvas class page. Instructor and TA office hours start in the second week of class. Most office hours will be online via Zoom, but some may be in person (you will be notified by email in advance). You are also welcome to talk to the instructor/TAs *in-person* right after class/tutorial. *You are strongly encouraged to post your questions on **Piazza discussion board** via Canvas class page.*

Teaching assistants:

See Canvas.

Lecture and course information:

The focuses of the lecture/course are (i) understanding of statistical models and methods (e.g., choosing the right statistical models/methods for a given dataset, which is very important in this AI age), (ii) **case studies** and real data analysis, and (iii) critical thinking in statistics. These skills are particularly important in the AI age. *All lectures are in-person.* Almost all lectures will involve either **iClicker questions** or **in-class activities** or both – these are designed for you to test your understanding of the course materials and for you to develop the foregoing mentioned skills.

Tutorial:

Tutorials start in the *second week* of class, in-person.

Course policy:

Regular attendance in lectures and tutorials is expected of each student. **You must bring a laptop to each lecture class for in-class activities, which will be counted**

for final grades (if you don't have a laptop, you should be able to borrow one from UBC Library). Late assignment submissions (in-class activities, lecture and tutorial worksheets, project) will not be accepted – the deadlines shown on Canvas are firm with no exceptions (no excuse). You are strongly encouraged to complete the lecture/tutorial worksheets and project early to avoid last minute issues.

Email policy:

Due to large volumes of emails the instructor receives daily, the instructor may be unable to respond individual emails on time. *Please use email for confidential and personal matters only.* Please use the online **Piazza discussion board** on Canvas to discuss course materials, or talk to the instructor/TA right after class/tutorial.

Textbook:

There is no required textbook for this course. The lecture notes posted on Canvas will define the course materials. Some **references** are listed at the end, which can be freely accessed online via the UBC library.

Prerequisite:

DSCI 100, STAT 201, and one of MATH 100, MATH 102, MATH 104, MATH 110, MATH 120, MATH 180, MATH 184, SCIE 001.

Assignments:

The assessment of learning will have five components (the time required to complete each component is not much, except the group project):

- *Lectures' Worksheets*: fully auto-graded with visible tests to help you identify points that need more clarification. Weekly assignments (each should take less than 20 minutes). The lowest two worksheet grades will be dropped.
- *Tutorials' Worksheets*: only a few exercises will have visible tests. Weekly assignments (each should take less than 20 minutes). The lowest two worksheet grades will be dropped.
- *Group project*: A project that you will work with your group throughout the term (you find your group members by yourselves). Details about the group project will be made available to you on Canvas soon.
- *iClicker*: We will be using iClicker in most lecture classes. You will be graded on *both participation and performance* (50% each). The two lowest iClicker marks will be dropped.
- *In-class activity*: There are designed for you to practice in most classes. **You must submit your activity work to Canvas by the end of each class** (i.e., 5:00pm on Tuesdays and Thursdays). These will be lightly graded – the key is participation in the activity (i.e., you will get the credits as long as you show that you have participated in the activity with some reasonable results even though some results may be incorrect).

Both lecture and tutorial worksheets are due
each Sunday by 11:59pm.

You should complete these assignments individually and independently. Please read the “Instructions for all worksheets and tutorials” on Canvas “Assignments” section for more details. No late submissions will be accepted (no excuses).

Since the summer classes are intensive, there will be **at least one** lecture worksheets and **at least one** tutorial worksheets **each week**. Please see Canvas Assignment section for more details. Worksheets and tutorial problem sets will be done using Jupyter Notebook to an external site and R. Students will access the worksheets and tutorials in Jupyter Notebooks through Canvas. *If you have any questions regarding lectures or tutorials’ worksheets, please post them on Piazza.* Do not email to instructor or TAs.

Group project:

There will be one group project, due on
Sunday, August 2, by 11:59pm

(upload on Canvas). Please see Canvas project description for details.

Piazza discussion board:

We will use the Piazza online platform (access via Canvas) to allow students, TAs, and the instructor to discuss course materials. In particular, if you have any questions regarding the assignments and course materials, please post them on Piazza. Do not email TAs or the instructor. For each question posted on Piazza, we will let other students to answer it first, unless the question has to be answered by instructor or TAs. The TAs and the instructor will be spending time moderating and posting to Piazza, if the questions cannot be answered by students. **Students are strongly encouraged to answer questions posted on Piazza** (answers by students will be checked by the TAs or instructor and may be revised if necessary). **The top 10% students who contribute the most on Piazza (i.e., actively answering questions on Piazza) will get 2 bonus points on their final grades** (e.g., your final grade will be 90% if your actual grade is 88%).

Class announcements:

Class announcements will be sent via emails. Please check your emails regularly.

R software:

We will use software R in class. R code will not be tested in the exams directly, but it is important to understand course materials illustrated in R (these will be tested in the exams). The data analysis in the group project has to be done in R.

Exams:

There will be one midterm exam and one final exam – both are **written exams in person.** In both exams, you will be asked to write down the key steps of your solutions, and then submit your solutions to Canvas by the end of the exams (so it’s important to understand the statistical models and methods). The midterm exam will be on

Thursday, July 23, in class (one hour).

Details of the exams can be found on Canvas. Final exam date/time will be announced by the university (final exam will be 2.5 hours). **No makeup exams.** If you miss a midterm with good reasons (e.g., sick), the midterm weight will be transferred to your final exam. If you miss the final exam with good reasons, you have to formally apply for a deferred final exam through your Faculty (e.g., Faculty of Science Advising Office). Please see Canvas for more details about exams.

Evaluation:

Your final grade will be computed based on the following weighting scheme: iClicker: 5%, in-class activity: 5%, lecture worksheets: 5%, tutorial worksheets: 5%, group project: 10%, midterm: 20%, final exam: 50%. The weights may be adjusted if necessary. You do not need to pass the final exam to pass the course.

Lecture topics:

1. Linear regression models (estimation, inference, assumptions, and diagnostics),
2. Logistic regression models,
3. Poisson regression models,
4. Model evaluation and goodness of fit,
5. Model regularization and post-inference,
6. Prediction uncertainty,
7. Classifiers,
8. A/B testing (if time permits).

References:

The following references are closely related to course materials and can be found on UBC Library:

- Introduction to statistical learning (R version) by Gareth James, Daniela Witten, Trevor Hastie, and Rob Tibshirani
- Introduction to Data Science by Rafael A. Irizarry
- Beyond Multiple Linear Regression by Paul Roback and Julie Legler